

Lecture 3. Deep Learning for Continuous Time Models and Structural Estimation

Yucheng Yang

University of Zurich and SFI; ERID Visitor, Duke University

Duke Mini-Course on Machine Learning for Dynamic Economic Models

September 2026

Background: deep learning for continuous-time models

- Lectures 1–2: discrete-time HA models with aggregate shocks; distribution as a state (“master equation”) is either solved directly (DeepHAM) or sidestepped (SRL)
- Many dynamic models are set in continuous time: HJB + KFE for HA models (Achdou et al '22), macro-finance (Brunnermeier-Sannikov), search and matching
- With heterogeneity + aggregate shocks, equilibrium = high-dimensional PDE
- Deep learning solves such PDEs globally: neural nets as function approximators, PDE residuals as loss, economics-guided sampling and training
- Furthermore, solve models with structural parameters as pseudo state variables $\Rightarrow$ deep learning based structural estimation
- This lecture is based on Payne, Rebei & Yang (2026), “Deep Learning for Search and Matching Models” (DeepSAM)

Introduction

- **Heterogeneity** and **aggregate shocks** are important in markets with **search frictions**. (e.g. labor and financial markets).
- Most search and matching (SAM) models with heterogeneous agents study:
 1. Deterministic steady state (e.g. Shimer-Smith '00),
 2. Aggregate fluctuations, but impose restrictions to eliminate distribution from state space (e.g. “block recursivity” in Menzio-Shi '11, Lise-Robin '17; Lagos-Rocheteau '09).
- We present SAM models as **high-dim. PDEs** with **distribution** & **agg. shocks** as states ...and develop a new deep learning method, DeepSAM, to **solve** them globally.
- We extend DeepSAM for **internal calibration with SMM** in efficient computation time.

DeepSAM: Payne, Rebei & Yang (2026)

- Develop DeepSAM and apply to “canonical” search models with aggregate shocks:
 1. Shimer-Smith/Mortensen-Pissarides model with two-sided heterogeneity.
 2. Lise-Robin on-the-job search (OJS) model with worker bargaining power.
 3. Duffie-Garleanu-Pederson OTC model with asset and investor heterogeneity.
- High accuracy + efficient compute time for both solution and estimation.
- This talk: study unemployment and wage dynamics during business cycles and crises:
 1. Distribution feedback most important when aggregate shocks are asymmetric.
 2. Low-type worker wages are more procyclical.
 3. Low-type workers benefit more from longer expansions (“Okun’s hypothesis”).
 4. Block recursive model assumptions amplify IRFs.

Literature

- Deep learning for incomplete-market heterogeneous agent models (HAM): [Lecture 1](#)
 - [DeepSAM: search and matching \(SAM\) models.](#)

	Distribution	Distribution impact on decisions
HAM	Asset wealth and income	Via aggregate prices
SAM	Type (productivity) of agents in two sides of matching	Via matching process with other types

- Search model with business cycle (e.g. Shimer '05, Menzio-Shi '11, Lise-Robin '17.)
 - [DeepSAM: keep distribution in the state vector.](#)
- Integrate deep learning based solution methods with calibration and estimation (e.g. Chen et al '26, Kase et al '26, Friedl et al '23)
 - [DeepSAM: standard internal calibration practice for quantitative macro.](#)

Table of Contents

Methodology

Application 1: Crisis Shock in Shimer-Smith (2000)

Application 2: Business Cycles in Lise-Robin (2017) With Worker Bargaining Power

Conclusion

Shimer-Smith Model with Aggregate Shocks

- Continuous time, infinite horizon environment.
- **Workers** $x \in [0, 1]$ with exog density $g_t^w(x)$; **Firms** $y \in [0, 1]$ with $g_t^f(y)$ by free entry:
 - Unmatched: unemployed workers get benefit b ; vacant firms pay vacancy cost c .
 - Matched: type x worker and type y firm produce output $z_t f(x, y)$.
 - z_t : follows continuous time Markov Chain (can be generalized).
 - Firms make entry decision and then draw a type y from uniform distribution $[0, 1]$. More
 - Firm exit: exogenous exit at rate $\varsigma(z)$; fraction $\sigma(z, \tilde{z})$ of firms exit when z jumps to $\tilde{z}$.
(Exiting firms/workers compensated; calibrated s.t. **no endogenous exit**.) General case
- **Meet randomly** at rate $m(\mathcal{U}_t, \mathcal{V}_t)$, $\mathcal{U}_t$ is total unemployment, $\mathcal{V}_t$ is total vacancies.
- Upon meeting, agents choose whether to accept the match:
 - Match surplus $S_t(x, y)$ divided by **Nash bargaining**: worker gets fraction β .
 - Acceptance decision $\alpha_t(x, y) = \mathbb{1}\{S_t(x, y) \geq 0\}$. Exogenous dissolve rate $\delta(x, y, z)$.
- Equilibrium object: $g_t(x, y)$ “density” of matches $\Rightarrow$ unemployed $g_t^u(x)$, vacant $g_t^v(y)$.

Recursive Equilibrium: Agent Problems

- E.g. worker idiosyncratic state = x , Aggregate states = $(z, g(x, y))$.
Let $dg/dt = \check{\mu}^g(x, y, z, g)$ denote agents' belief about the evolution of g .
- Hamilton-Jacobi-Bellman equation (HJBE) for unemployed worker's value $V^u(x, z, g)$:

$$\begin{aligned} \rho V^u(x, z, g) = & b + \frac{m(z, g)}{U(z, g)} \int \underbrace{\overbrace{\alpha(x, \tilde{y}, z, g)}^{\text{acceptance decision}} \overbrace{(V^e(x, \tilde{y}, z, g) - V^u(x, z, g))}^{\text{employed value}}}_{\text{change of value conditional on match}} \frac{g^v(\tilde{y})}{V(z, g)} d\tilde{y} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (V^u(x, \tilde{z}, g^+) - V^u(x, z, g)) + \underbrace{D_g V^u(x, z, g)}_{\text{Frechet derivative: how change of } g \text{ affects } V} \cdot \underbrace{\check{\mu}^g}_{\text{Belief about } g \text{ evolution}} \end{aligned}$$

where $g^+ := (1 - \sigma(z, \tilde{z}))g$: post-jump distribution (firm exit at z jumps).

Recursive Equilibrium: Agent Problems

- E.g. worker idiosyncratic state = x , Aggregate states = $(z, g(x, y))$.
Let $dg/dt = \check{\mu}^g(x, y, z, g)$ denote agents' belief about the evolution of g .
- Hamilton-Jacobi-Bellman equation (HJBE) for unemployed worker's value $V^u(x, z, g)$:

$$\begin{aligned} \rho V^u(x, z, g) = & b + \frac{m(z, g)}{U(z, g)} \int \underbrace{\overbrace{\alpha(x, \tilde{y}, z, g)}^{\text{acceptance decision}} \overbrace{(V^e(x, \tilde{y}, z, g) - V^u(x, z, g))}^{\text{employed value}}}_{\text{change of value conditional on match}} \frac{g^v(\tilde{y})}{V^v(z, g)} d\tilde{y} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (V^u(x, \tilde{z}, g^+) - V^u(x, z, g)) + \underbrace{D_g V^u(x, z, g)}_{\text{Frechet derivative: how change of } g \text{ affects } V} \cdot \underbrace{\check{\mu}^g}_{\text{Belief about } g \text{ evolution}} \end{aligned}$$

where $g^+ := (1 - \sigma(z, \tilde{z}))g$: post-jump distribution (firm exit at z jumps).

+ HJBEs for employed workers V^e , vacant firms V^v , and producing firms V^p More.

Recursive Equilibrium: Distribution Evolution and Definition

Given α choices, **evolution of $g_t(x, y)$** is given by Kolmogorov forward equation (KFE):

$$dg_t(x, y) = \mu^g(x, y, z_t, g_t) dt - \underbrace{\sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z})}_{\text{Jump: exogenous firm exit when } z_t \text{ jumps to } \tilde{z}}$$

$$\mu^g(x, y, z, g) = \underbrace{-(\delta(x, y, z) + s(z)) g_t(x, y)}_{\text{Outflow: breakup of matches \& firm exit}} + \underbrace{\frac{m(z, g)}{\mathcal{U}(z, g)\mathcal{V}(z, g)} \alpha(x, y, z, g) g_t^u(x) g_t^v(y)}_{\text{Inflow: creation of new matches}}$$

Recursive Equilibrium: Distribution Evolution and Definition

Given α choices, **evolution of $g_t(x, y)$** is given by Kolmogorov forward equation (KFE):

$$dg_t(x, y) = \mu^g(x, y, z_t, g_t) dt - \underbrace{\sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z})}_{\text{Jump: exogenous firm exit when } z_t \text{ jumps to } \tilde{z}}$$

$$\mu^g(x, y, z, g) = \underbrace{-(\delta(x, y, z) + s(z)) g_t(x, y)}_{\text{Outflow: breakup of matches \& firm exit}} + \underbrace{\frac{m(z, g)}{\mathcal{U}(z, g)\mathcal{V}(z, g)} \alpha(x, y, z, g) g_t^u(x) g_t^v(y)}_{\text{Inflow: creation of new matches}}$$

A (recursive) equilibrium is functions $\{V^u, V^e, V^v, V^p, w, \alpha, g^f\}$ of state variables (z, g) such that:

1. Given beliefs about the evolution of g_t , $(V^u, V^e, V^v, V^p, \alpha)$ solve the HJB equations,
2. The division of surplus satisfies Nash bargaining,
3. g^f satisfies the free entry condition details, and
4. Agent beliefs about the evolution of g_t are consistent: $\check{\mu}^g = \mu^g$.

Recursive Characterization For Equilibrium Surplus

- Surplus of match $S(x, y, z, g) \equiv V^p(x, y, z, g) - V^v(y, z, g) + V^e(x, y, z, g) - V^u(x, z, g)$
- We characterize equilibrium with master equation for surplus: C.f. block recursive

$$\begin{aligned} \rho S(x, y, z, g) = & z f(x, y) - \delta(x, y, z) S(x, y, z, g) \\ & + c - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g; S)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\ & - b - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g; S)} d\tilde{y} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (S(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - S(x, y, z, g)) + D_g S \cdot \mu^g(z, g) \end{aligned}$$

- Kolmogorov forward equation (KFE): $dg_t(x, y) = \mu^g dt - \sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z}),$

$$\mu^g(x, y, z, g) = -(\delta(x, y, z) + \varsigma(z)) g(x, y) + \frac{m(z, g)}{\mathcal{U}(z, g) \mathcal{V}(z, g)} \alpha(x, y, z, g) g^v(y) g^u(x)$$

- High-dim PDEs with distribution in state: hard to solve with conventional methods.

Finite Type Approximation

- Approximate $g(x, y)$ on finite types: $x \in \mathcal{X} = \{x_1, \dots, x_{n_x}\}$, $y \in \mathcal{Y} = \{y_1, \dots, y_{n_y}\}$.
(Other approaches of discretization: projection and finite population [More](#))
- Finite state approximation $\Rightarrow$ analytical (approximate) KFE: $g \approx \{g_{ij}\}_{i \leq n_x, j \leq n_y}$
- Approximated master equation for surplus:

$$0 = \mathcal{L}^S S(x, y, z, g) = -(\rho + \delta(x, y, z))S(x, y, z, g) + zf(x, y) + c - b$$

$$- (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g)} \frac{1}{n_x} \sum_{i=1}^{n_x} \alpha(\tilde{x}_i, y, z, g) S(\tilde{x}_i, y, z, g) \frac{g^u(\tilde{x}_i)}{\mathcal{U}(z, g)}$$

$$- \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \frac{1}{n_y} \sum_{j=1}^{n_y} \alpha(x, \tilde{y}_j, z, g) S(x, \tilde{y}_j, z, g) \frac{g^v(\tilde{y}_j)}{\mathcal{V}(z, g)}$$

$$+ \sum_{\tilde{z}} \lambda_{z\tilde{z}} (S(x, y, \tilde{z}, g^+) - S(x, y, z, g)) + \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \partial_{g_{ij}} S(x, y, z, \{g_{ij}\}_{i,j}) \mu^g(\tilde{x}_i, \tilde{y}_j, z, g)$$

where $g_{ij}^+ = (1 - \sigma(z, \tilde{z}))g_{ij}$; μ^g from discretized KFE.

Discrete Choice Approximation

- In the original model, acceptance is discrete: $\alpha(x, y, z, g) = \mathbb{1}\{S(x, y, z, g) \geq 0\}$
- Discrete choice $\alpha \Rightarrow S(x, y, z, g)$ non-smooth at some g (a viscosity solution)
 $\Rightarrow$ sharp decision boundaries make gradient-based NN training less stable.
- Baseline: keep master equation well defined & NN training efficient $\Rightarrow$ approximate:

$$\alpha(x, y, z, g) = \frac{1}{1 + e^{-\xi S(x, y, z, g)}}$$

Interpretation: logit choice model with utility shocks $\sim$ extreme value distribution
(micro-foundation as in Caliendo et al. '19; $\xi \rightarrow \infty \Rightarrow$ discrete choice α).

- We can also solve the original discrete choice formulation, via “homotopy”:
 1. Train on a sequence of smooth problems with increasing curvature $\xi_1 < \dots < \xi_N$, warm-starting each stage from the previous solution;
 2. Final stage: switch to the exact discrete α , initialized from the ξ_N solution.
- Delivers accurate solutions to the discrete choice problem ($\xi = 42$ already $\approx$ discrete),
...but $\sim 10\times$ compute time: smooth α preferable if logit interpretation acceptable.

Results

DeepSAM Algorithm for Solving the Master Equation

1. Approximate surplus by neural network $S(x, y, z, g) \approx \hat{S}(x, y, z, g; \Theta)$. Function form
2. Start with initial parameter guess Θ^0 . At iteration n with Θ^n :
 - 2.1 Generate K sample points, $Q^n = \{(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y})\}_{k \leq K}$.
 - 2.2 Calculate the average mean squared error of surplus master equation on sample points:

$$L(\Theta^n, Q^n) := \frac{1}{K} \sum_{k \leq K} \left| \mathcal{L}^S \hat{S}(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}) \right|^2$$

- 2.3 Update NN parameters with stochastic gradient descent (SGD) or variants:

$$\Theta^{n+1} = \Theta^n - \zeta^n \nabla_{\Theta} L(\Theta^n, Q^n)$$

- 2.4 Repeat until $L(\Theta^n, Q^n) \leq \epsilon$ with precision threshold ϵ .

3. Once S is solved, we have α and can solve for worker and firm value functions.

Tutorial (Colab notebooks): [Tutorials/Tutorial_DeepSAM in the course repository](#)

Comment: Some Technical Details

- Q. How do we choose where to sample? Three-phase sampling & training scheme:
 - **Phase 1**: draw distributions “between” steady states (random mixtures of steady states for different fixed z); train until the loss is small.
 - **Phase 2**: move to simulation based (“ergodic”) sampling under the current $\hat{S}$, with a training/validation split, and periodic resimulation as $\hat{S}$ improves.
 - **Phase 3**: report out-of-sample loss on a freshly simulated evaluation set. [More](#)
- Q. Why global solution? Non-linear approximation across broad state space.
- Q. Which equilibrium concept?
 - We solve for a rational expectations equilibrium
 $\Rightarrow$ requires solving the master equation where $\mu^g = \check{\mu}^g$ on and off equilibrium path.
 - Different to “reinforcement learning” where agents optimize in response to simulations
 $\Rightarrow \mu^g \neq \check{\mu}^g$ does not necessarily hold, especially off equilibrium path. [More](#)

Neural networks do not train in seconds.

So how can we choose structural parameters to match data?

DeepSAM for Internal Calibration with Simulated Method of Moments

- So far, we have solved:

$$0 = \mathcal{L}^S S(x, y, z, \underline{g})$$

- For internal calibration, we include economic parameters Ψ as pseudo states and solve extended master equation:

$$0 = \mathcal{L}^S S(x, y, z, \underline{g}, \Psi)$$

- Simulate the model under different structural parameter vectors $\{\Psi_I\}_I$ and ...fit a surrogate function mapping structural parameters to simulated moments.
- Use surrogate function to find the parameters that match data moments.

Extensions

- On-the-job search with a parameterized bargaining protocol nesting: [More](#)
 - Bertrand competition between firms (Postel-Vinay and Robin, 2002),
 - sequential auctions (Cahuc, Postel-Vinay and Robin, 2006), and
 - Nash bargaining over incremental surplus from moving (Pissarides, 1994).
- Match specific idiosyncratic productivity shocks. [More](#)
- Endogenous separation. [More](#)
- Variations on free entry and exit. [More](#)
- Agent type switching. [More](#)
- Asset trade rather than worker-firm production (e.g., OTC search markets). [More](#)
- Non-transferable utility. [More](#)

Table of Contents

Methodology

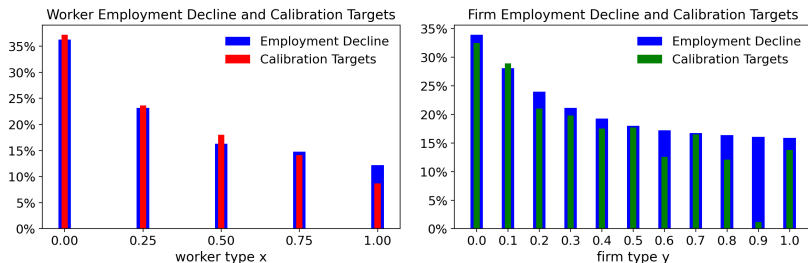
Application 1: Crisis Shock in Shimer-Smith (2000)

Application 2: Business Cycles in Lise-Robin (2017) With Worker Bargaining Power

Conclusion

Crisis Shock in Shimer-Smith (2000)

- We test the solution method for the baseline model
...with a recession where the job separation rate significantly increases.
- We calibrate the crisis separation rate $\delta_D(x, y)$ to match the heterogeneous employment declines during COVID-19 for different workers/firms (Cajner et al., 2020). [More](#)



- Worker and firm types: $(n_x, n_y) = (5, 11) \Rightarrow 58$ dimensional PDE for $S(x, y, z, g)$

Numerical Accuracy of Solution Technique

Technically, our method achieves high numerical accuracy using a number of measures:

- **Small numerical error.** Use DeepSAM to solve the problem (58 dimensional PDE), and compute loss on simulated paths not used in training. Evaluation RMSE: 1.7×10^{-3} (training RMSE: 9.2×10^{-4}). [More](#)
- **Verification on models with known solution.** Use DeepSAM to solve model without aggregate shocks (57 dimensional PDE) and obtain steady state solution. Compare to steady state solution from conventional methods. RMSE Difference: 2.4×10^{-3} . [More](#)

Q1. Does Distribution Feedback Matter? Evidence From COVID-19

- Aggregate dynamics **with** and **without** distribution feedback to agent decisions:

Full dynamics: $\mu_t^g(x, y) = -(\delta(x, y, z_t) + \varsigma(z_t))g_t(x, y) + \frac{m_t(z, g_t)}{\mathcal{U}_t \mathcal{V}_t} \alpha(x, y, z_t, g_t) g_t^u(x) g_t^v(y)$

No distribution feedback: $\mu_t^g(x, y) = -(\delta(x, y, z_t) + \varsigma(z_t))g_t(x, y) + \frac{m_t(z, g_t)}{\mathcal{U}_t \mathcal{V}_t} \alpha(x, y, z_t, g^{\text{ergodic}}) g_t^u(x) g_t^v(y)$

(Both share the same exogenous exit jumps at z_t changes; g^{ergodic} : ergodic mean of g_t .)

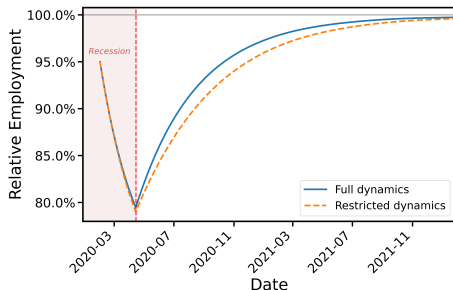
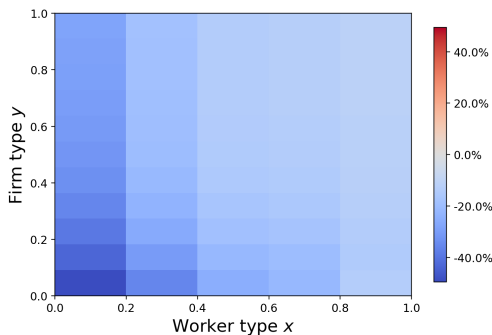


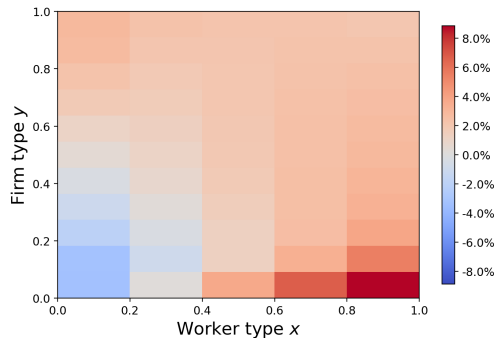
Figure: Employment change after the COVID-19 shock

A1. Mechanism: Asymmetric Shock Makes Agents “Less Picky”

Intuition: COVID-19 leads to relatively more low-type unemployed workers and firms
⇒ workers and firms are less willing to wait for a good match.



(a) Distribution difference: after COVID-19 shock relative to the ergodic mean.



(b) Acceptance difference: after COVID-19 shock relative to the ergodic mean.

Table of Contents

Methodology

Application 1: Crisis Shock in Shimer-Smith (2000)

Application 2: Business Cycles in Lise-Robin (2017) With Worker Bargaining Power

Conclusion

Solving and Estimating OJS Model

- Lise-Robin (2017) with worker bargaining power $\beta > 0$ [Model details](#)
- Discretization: $(n_x, n_y) = (7, 8) \Rightarrow S(x, y, z, g)$ is 59-dimensional.

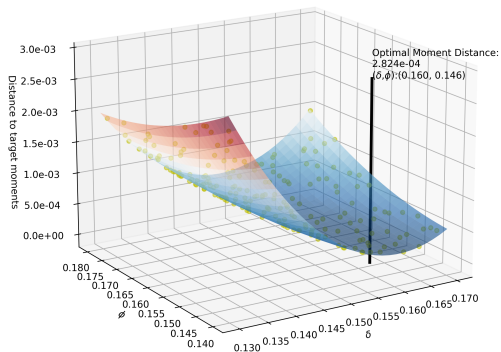
Solving and Estimating OJS Model

- Lise-Robin (2017) with worker bargaining power $\beta > 0$ Model details
- Discretization: $(n_x, n_y) = (7, 8) \Rightarrow S(x, y, z, g)$ is 59-dimensional.
- Internal calibration $\{\beta, \kappa, c, \delta, \phi, \eta, b\}$: solve model over economic parameter space, simulate across 10,000 parameter combinations for simulated method of moments.

	Solving Model For a Parameter Set	Solving Model with Structural Parameters as Pseudo-states	Training Surrogate Model	Simulated Method of Moments	Entire Estimation
RMSE Loss	9.97×10^{-4}	2.9×10^{-3}	7.82×10^{-4}	1.24×10^{-4}	-
Time	40min	2h 47min	23min	0.4min	3h 10min

Moments	$\mathbb{E}[U]$	$\mathbb{E}[V]$	$\mathbb{E}[EE]$	$\mathbb{E}[UE]$	$\mathbb{E}[EU]$
Data	0.058	0.037	0.025	0.421	0.025
Model	0.057	0.036	0.026	0.448	0.027

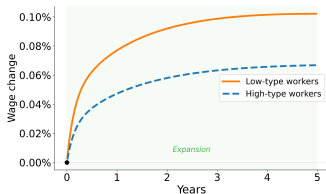
Estimation of OJS Model: Visualization in 2D



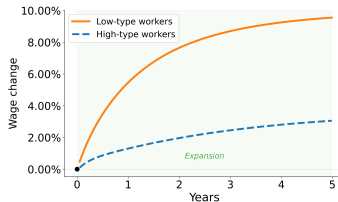
Distance to target moments over separation rate δ and OJS intensity ϕ .

Q2. Are wage dynamics heterogeneous across distribution? A2. Yes

- In Lise-Robin: “wages cannot be solved for exactly... need to solve worker values where the distribution of workers across jobs is a state variable.”
- DeepSAM can solve wage dynamics with rich heterogeneity.
- Low-type worker wages more procyclical — under both protocols $\chi = 0$ and $\chi = 1$.



(a) IRF to positive shocks ($\chi = 0$)



(b) IRF to positive shocks ($\chi = 1$)

Figure: Wage change after z shocks for $\chi = 0$, $\chi = 1$. Low/middle/high: lowest, middle, highest of $n_x = 7$ worker types.

Q3. Who benefits more over a longer expansion? A3. Low Types

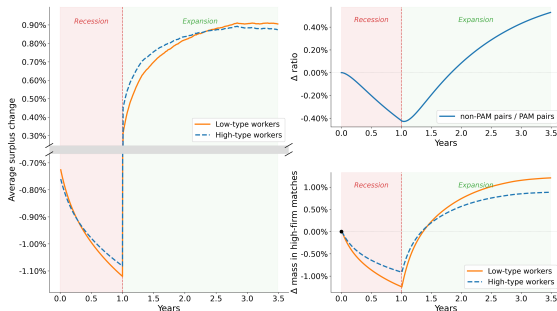


Figure: Left: Δ avg. surplus by worker type. Top right: non-PAM/PAM mass ratio. Bottom right: Δ employment at firms with $y > 1/2$.

- Low-type workers overtake high types in surplus growth ~ 2 years into the expansion.
- Mechanism: sorting weakens over time in expansions — high-type workers become scarce, so high-type firms become less picky and hire more low-type workers.

Table of Contents

Methodology

Application 1: Crisis Shock in Shimer-Smith (2000)

Application 2: Business Cycles in Lise-Robin (2017) With Worker Bargaining Power

Conclusion

Conclusion

- We develop an integrated global solution and estimation method, DeepSAM, to search and matching models with heterogeneity and aggregate shocks.
- We apply DeepSAM to three general setups in labor and financial search models (without simplification assumptions). [OTC model details](#)
- The method is accurate, solves new variables (e.g. wage), and generates novel economic insights.
- A foundational tool for a large literature with more applications:
 - Richer models in labor, financial, and money search, combined with rich micro data.
 - Spatial, trade, and network models with aggregate uncertainty (similar math structure).

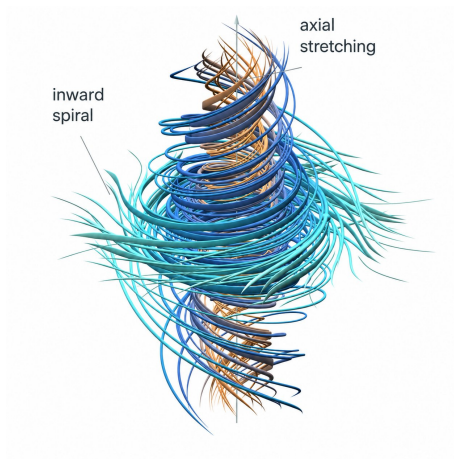
The Agenda of “AI for Economics”: Two Generations of Work

- First generation: “**proof of concept**” of using AI to solve economic models.
 - Duarte (2018), Azinovic, Gaegauf, Scheidegger (2019), Maliar, Maliar, Winant (2019), Fernández-Villaverde, Hurtado, Nuño (2019), Han, Yang, E (2021), Kahou, Fernández-Villaverde, Perla, Sood (2021), Valaitis and Villa (2021), Gopalakrishna (2022), Huang (2022), Gu, Lauriere, Merkel, Payne (2023), Zhong (2023), among others.
 - All papers solve similar **classical models** in this generation.
- Second generation: apply AI to **long-standing difficult questions in various literature**.
 - Climate economics: Folini, Friedl, Kubler, Scheidegger (2021), Friedl, Kubler, Scheidegger, Usui (2023), Barnett, Brock, Hansen, Hu, Huang (2023), Sun (2023).
 - Labor and financial markets: Payne, Rebei, and Yang (2026) $\Rightarrow$ Lecture 3.
 - Asset pricing: Zhang (2022), Azinovic-Zemlicka (2023), Gopalakrishna, Gu, Payne (2024).
 - HANK models: F-V, Marbet, Nuño, Rachedi (2022), Kase, Melosi, Rottner (2022).
 - Network models: Carvalho, Covarrubias, Nuño (2025).
 - Many other fields: trade, optimal policy, information, structural estimation, etc.

Summary

- Three elements of using deep learning to solve economic models:
 1. Parameterize the functions of interest as neural networks, can impose economic restrictions in parameterization
 2. Specify the objective
 3. Choose where to evaluate the objective (sampling)
- As is happening in other disciplines ([Nobel Prize Physics 2024](#), [Chemistry 2024](#), [Navier-Stokes Solution 2026](#)), AI may become the new generation of model solvers in economics.

Thank You for a Wonderful Week!



Approximate S by Neural Network (Feed Forward, Fully Connected)

- Let $\omega = (x, y, z, g)$. We approximate surplus $S(\omega)$ by neural network with form:

$$\mathbf{h}^{(1)} = \phi^{(1)}(W^{(1)}\omega + \mathbf{b}^{(1)}) \quad \dots \text{Hidden layer 1}$$

$$\mathbf{h}^{(2)} = \phi^{(2)}(W^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}) \quad \dots \text{Hidden layer 2}$$

$$\vdots$$

$$\mathbf{h}^{(H)} = \phi^{(H)}(W^{(H)}\mathbf{h}^{(H-1)} + \mathbf{b}^{(H)}) \quad \dots \text{Hidden layer H}$$

$$S = \sigma(\mathbf{h}^{(H)}) \quad \dots \text{Surplus}$$

- Terminology (our parameter choices are in blue):
 - H : is the number of hidden layers, ($H = 4$; 5 for estimation with pseudo-states)
 - Length of vector $\mathbf{h}^{(i)}$: number of neurons in hidden layer i , ($Length = 50$; 100 for estimation)
 - $\phi^{(i)}$: is the activation function for hidden layer i , ($\phi^i = \tanh$)
 - σ : is the activation function for the final layer, ($\sigma = \tanh$)
 - $\Theta = (W^1, \dots, W^{(H)}, b^{(1)}, \dots, b^{(H)})$ are the parameters,

Table of Contents

Methodology

Labor Search Model

Entry and Exit

On-The-Job Search Model

On-The-Job Search Model with Idiosyncratic Shocks

OTC Market

Non-transferable Utility (NTU)

Free Entry Condition

- Firms make entry decision and then draw type y from uniform distribution $[0, 1]$:

$$0 = \mathbb{E}[V_t^v] = \int V^v(\tilde{y}, z, g) d\tilde{y}. \quad (1)$$

- As the matching function is homothetic $\frac{m(z_t, g_t)}{\mathcal{V}_t} = \hat{m}\left(\frac{\mathcal{V}_t}{\mathcal{U}_t}\right)$, combining free entry condition with HJB equation for V^v gives:

$$\hat{m}\left(\frac{\mathcal{V}_t}{\mathcal{U}_t}\right) = \frac{c}{\int \int \alpha(\tilde{x}, \tilde{y}) \frac{g_t^u(\tilde{x})}{\mathcal{U}_t} (1 - \beta) S_t(\tilde{x}, \tilde{y}) d\tilde{x} d\tilde{y}} \Rightarrow \mathcal{V}_t = \mathcal{U}_t \hat{m}^{-1}(\dots) \quad (2)$$

where $g_t^u = g_t^w - \int g_t^m(x, y) dy$ and so the RHS can be computed from g_t^m and S_t .

- $g_t^f(y) = \mathcal{V}_t + \mathcal{P}_t$ for all y (uniform), where $\mathcal{V}_t, \mathcal{P}_t$ can be expressed in terms of g and S .
- Timing at a jump in z : z changes $\rightarrow$ exogenous firm exit (σ) $\rightarrow g_t^f$ adjusts to restore free entry.
- With free entry condition, the master equation expression for surplus takes the same form as without free entry, but with different expressions of $g^f(y)$.

Recursive Equilibrium Part II: Other Equations

- Hamilton-Jacobi-Bellman equation (HJBE) for employed worker's value $V^e(x, y, z, g)$:

$$\begin{aligned}\rho V^e(x, y, z, g) = & w(x, y, z, g) + \delta(x, y, z) (V^u(x, z, g) - V^e(x, y, z, g)) \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^e(x, y, \tilde{z}, g^+) - V^e(x, y, z, g)) + D_g V^e(x, y, z, g) \cdot \check{\mu}^g\end{aligned}$$

- HJBE for a vacant firm's value $V^v(y, z, g)$:

$$\begin{aligned}\rho V^v(y, z, g) = & -c + \frac{m(z, g)}{\mathcal{V}(z, g)} \int \alpha(\tilde{x}, y, z, g) (V^p(\tilde{x}, y, z, g) - V^v(y, z, g)) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^v(y, \tilde{z}, g^+) - V^v(y, z, g)) + D_g V^v(y, z, g) \cdot \check{\mu}^g\end{aligned}$$

- HJBE for a producing firm's value $V^p(x, y, z, g)$:

$$\begin{aligned}\rho V^p(x, y, z, g) = & zf(x, y) - w(x, y, z, g) + \delta(x, y, z) (V^v(y, z, g) - V^p(x, y, z, g)) \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^p(x, y, \tilde{z}, g^+) - V^p(x, y, z, g)) + D_g V^p(x, y, z, g) \cdot \check{\mu}^g\end{aligned}$$

- $g^+ := (1 - \sigma(z, \tilde{z}))g$: post-jump distribution. No direct ς/σ terms: full compensation for exit.

Comparison to Other Heterogeneous Agent Search Models

- [Lise-Robin '17](#): sets $\beta = 0$ (and other conditions, including Postal-Vinay Robin style Bertrand competition for workers searching on-the-job)

$$S(x, y, z, g) = S(x, y, z), \quad \alpha(x, y, z, g) = \alpha(x, y, z)$$

- [Menzio-Shi '11](#): competitive search (directed across a collection of sub-markets):

$$S(x, y, z, g) = S(x, y, z)$$

- We look for a solution for S and α in terms of the distribution g .

Modification 1: Finite Type Approximation

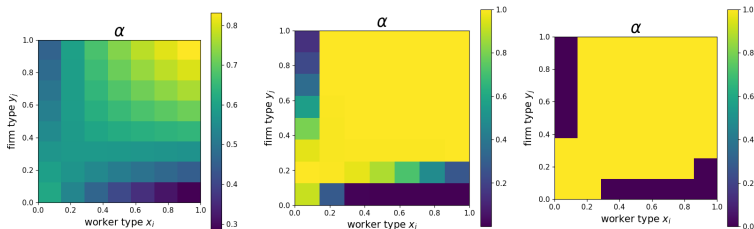
- Approximate $g(x, y)$ on finite types: $x \in \mathcal{X} = \{x_1, \dots, x_{n_x}\}$, $y \in \mathcal{Y} = \{y_1, \dots, y_{n_y}\}$.
- Finite state approximation $\Rightarrow$ analytical (approximate) KFE: $g \approx \{g_{ij}\}_{i \leq n_x, j \leq n_y}$
- Approximated master equation for surplus:

$$\begin{aligned} 0 = \mathcal{L}^S S(x, y, z, g) = & -(\rho + \delta(x, y, z))S(x, y, z, g) + zf(x, y) + c - b \\ & - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g)} \frac{1}{n_x} \sum_{i=1}^{n_x} \alpha(\tilde{x}_i, y, z, g) S(\tilde{x}_i, y, z, g) \frac{g^u(\tilde{x}_i)}{\mathcal{U}(z, g)} \\ & - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \frac{1}{n_y} \sum_{j=1}^{n_y} \alpha(x, \tilde{y}_j, z, g) S(x, \tilde{y}_j, z, g) \frac{g^v(\tilde{y}_j)}{\mathcal{V}(z, g)} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (S(x, y, \tilde{z}, g^+) - S(x, y, z, g)) + \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \partial_{g_{ij}} S(x, y, z, \{g_{ij}\}_{i,j}) \mu^g(\tilde{x}_i, \tilde{y}_j, z, g) \end{aligned}$$

where $g_{ij}^+ = (1 - \sigma(z, \tilde{z}))g_{ij}$ is the post-jump match distribution.

Modification 2: Solving the Exact Discrete Choice by Homotopy

- Acceptance α at steady state g : smooth ($\xi = 2$, $\xi = 42$) vs. exact discrete rule:



- High curvature ($\xi = 42$) already closely approximates the exact discrete solution.
- Homotopy training is stable: loss declines within each ξ stage, jumps at stage switches, and the final discrete stage converges from the high- ξ warm start.
- C.f. perturbation approaches (Reiter '09, Bilal '23), which rely on differentiability.

Finite Dimensional “Distribution” Approximation ($a = \text{idio. state}$)

	Finite Population	Discrete State	Projection
Params $\hat{\varphi}$	Agent states $\hat{\varphi}_t = \{(a_t^i)\}_{i \leq N}$	Masses on grid $\hat{\varphi}_{i,t}, \forall (a^i)_{i \leq N}$	Basis coefficients $\hat{\varphi}_{i,t}, \forall b_i(a) _{i \leq N}$
Dist. approx.	$\frac{1}{N} \sum_{i=1}^N \delta_{(a_t^i)}$	$\sum_{i=1}^N \hat{\varphi}_{i,t} \delta_{(a^i)}$	$\sum_{i=0}^N \hat{\varphi}_{i,t} b_i(a)$ $\approx \mu_g \left(a, z, \sum_{i=1}^N \hat{\varphi}_{i,t} b_i \right)$
KFE approx. ($\mu^{\hat{\varphi}}$)	Evolution of agents' states	Evolution of mass between grid points (e.g. finite diff.)	Evolution of projection coefficients (least squares)

Finite population works well for Walrasian markets; less well for search [Back](#)

Sampling and Training Scheme for the Surplus Network

- Sampling (x, y, z) is easy (low-dim): multinomial sampler on the discretized support. Sampling the distribution (g_{ij}) is high-dim $\Rightarrow$ three-phase scheme:
- **Phase 1 — mixed steady-state sampler:**
 1. Solve for the steady state g_m^s for each fixed aggregate state $s \in \{L, H, D\}$.
 2. Draw random convex combinations $g = \sum_s w_s g_m^s$: draw a dominant state s^* with stationary prob. π_{s^*} , set $w_{s^*} \sim \text{Beta}(\alpha, 1)$ with $\alpha \gg 1$, split the rest $\propto \pi$.
 3. Train on these samples until the master equation loss falls below a threshold.
- **Phase 2 — ergodic simulation sampler, inner–outer training:**
 1. Simulate (z_t, g_t) forward with the KFE under the current $\hat{S}$; discard burn-in $\Rightarrow$ ergodic pool of (g, z) draws, split into training/validation sets $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}$.
 2. Inner loop: SGD on minibatches from $\mathcal{D}_{\text{train}}$; track validation loss on $\mathcal{D}_{\text{val}}$ and keep the best weights Θ^* (validation-based model selection).
 3. Outer loop: resimulate under $\Theta^* \Rightarrow$ fresh $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}$; repeat until validation loss stabilizes.
- **Phase 3 — out-of-sample evaluation:** simulate a new evaluation set $\mathcal{D}_{\text{eval}}$ after training; report the evaluation loss (the RMSE in our accuracy results).

Comment 2: Different NN Techniques $\Leftrightarrow$ Different Eqm Concepts

Borrowing from Sargent (2008): [Back](#)

- Rational expectations eqm ([this paper](#); $\mu^g = \check{\mu}^g$):
 - Agent “perceived law of motion (PLM)” is the true law of motion if agents follow PLM
 - True on and off the equilibrium path $\Rightarrow$ requires solving the master equation.
- Self-confirming eqm ([some model-based reinforcement learning](#), e.g. DeepHAM)
 - PLM is best statistical fit to data generated by economy where agents follow that PLM.
 - So, agents would never reject a misspecification test
 - but may have off-equilibrium beliefs that differ from rational expectations ($\mu^g \neq \check{\mu}^g$)
 - $\Rightarrow$ can be solved by updating reaction functions.
- Misspecified belief eqm
 - Agents have a “perceived law of motion” for the equilibrium variables in the economy
 - ...that satisfies a fixed parametric form but doesn't satisfy a misspecification test.

Calibration and Estimation Challenges

- Let $\Psi \in \Omega^\Psi$ be the structural parameters to be calibrated internally.
- Let $\check{\varphi} = (\check{\varphi}_1, \dots, \check{\varphi}_N)$ be the $N \times 1$ data moments that we want to match.
- Let $\phi(\Psi) = (\phi_1(\Psi), \dots, \phi_N(\Psi))$ be the corresponding simulated model moments.
- Computation challenge: need to solve $\phi(\Psi)$ for many structural parameters Ψ .
- Solution: include Ψ as a pseudo state vector and solve extended master equation:

$$0 = \mathcal{L}^S S(x, y, z, \underline{g}, \Psi)$$

- Approximate extended surplus function by NN: $S(x, y, z, g, \Psi) \approx \hat{S}(x, y, z, g, \Psi; \Theta)$.
- Fit $\hat{S}$ using same approach as before but sampling over states & structural parameters.

Simulated Method of Moments

- Fit surrogate neural network $\hat{\Phi}$ mapping structural parameters to simulated moments:

$$\Psi \mapsto \hat{\Phi}(\Psi; \Theta^\Phi) = (\hat{\phi}_1(\Psi; \Theta^\Phi), \dots, \hat{\phi}_N(\Psi; \Theta^\Phi)), \quad \Theta^\Phi \text{ are NN parameters}$$

- Simulate the model under many different structural parameter vectors $\{\Psi_I\}_I$
 - Compute the resulting model moments $\{(\phi_1(\Psi_I), \dots, \phi_N(\Psi_I))\}_I$
 - Fit neural network parameters Θ^Φ to approx relationship $\hat{\Phi}(\Psi; \Theta^\Phi)$ using simulations.
- Think: surrogate NN has incorporated the economic structure implicitly into Θ^Φ
 - With $\hat{\Phi}(\Psi; \Theta^\Phi)$, we can find the parameters to match data moments:
(i.e. perform simulated method of moments)

$$\Psi^* = \arg \min_{\Psi} \sum_{i=1}^N \omega_i \left(\frac{\check{\varphi}_i - \hat{\phi}_i(\Psi; \Theta^\Phi)}{\check{\varphi}_i} \right)^2.$$

Calibration and Estimation

1. Fit extended surplus function by NN: $S(x, y, z, g, \Psi) \approx \hat{S}(x, y, z, g, \Psi; \Theta)$.

Start with initial parameter guess Θ^0 . At iteration n with Θ^n :

- 1.1 Generate K sample points, $Q^n = \{(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}, \Psi_k)\}_{k \leq K}$.

- 1.2 Calculate the average mean squared error of surplus master equation on sample points:

$$L(\Theta^n, Q^n) := \frac{1}{K} \sum_{k \leq K} \left| \mathcal{L}^S \hat{S}(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}, \Psi_k) \right|^2$$

- 1.3 Update NN parameters with stochastic gradient descent (SGD) method.

2. Fit surrogate NN mapping structural parameters to moments: $\hat{\Phi}(\Psi; \Theta^\Phi)$.

3. Use surrogate NN to find parameters that match data moments.

Table of Contents

Methodology

Labor Search Model

Entry and Exit

On-The-Job Search Model

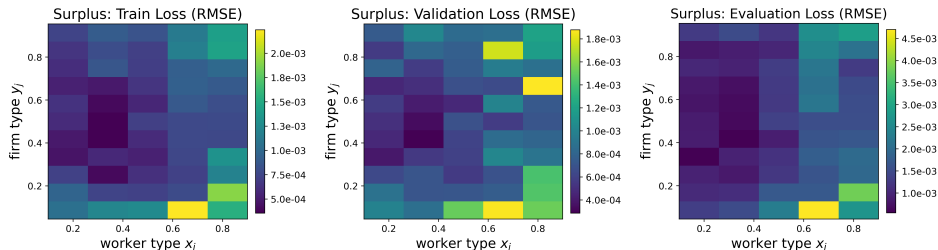
On-The-Job Search Model with Idiosyncratic Shocks

OTC Market

Non-transferable Utility (NTU)

Numerical Performance: Accuracy I [back](#)

- RMSE of the surplus master equation over the ergodic mean, computed on: (a) training samples (last inner loop), (b) validation samples (last outer loop), (c) a fresh simulated evaluation set.



- Average RMSEs: 9.2×10^{-4} (training), 9.1×10^{-4} (validation), 1.7×10^{-3} (evaluation)
 $\Rightarrow$ accurate approximation of S in the neighborhood of the ergodic mean.

Numerical Performance: Accuracy II

- Compare **steady state solution without aggregate shocks** to solution using conventional methods.

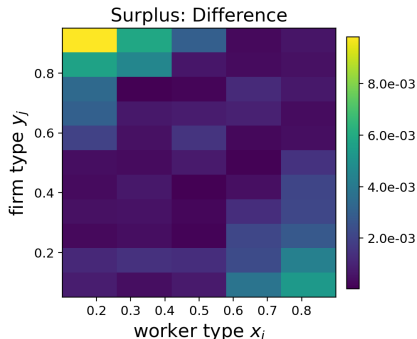
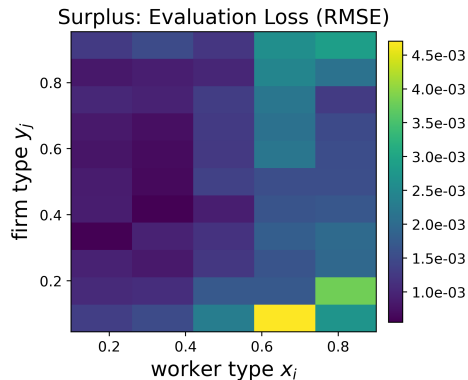


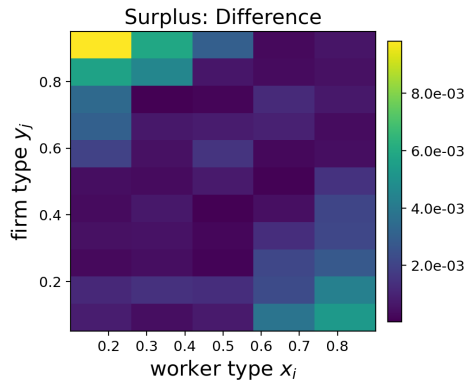
Figure: Absolute difference of S from conventional solution at DSS (order 10^{-3} ; max $S \approx 0.65$).

Calibration of Model with Aggregate Shocks Back

Parameter	Interpretation	Value	Target/Source
ρ	Discount rate	0.05	Interest rate
ξ	Extreme value for α choice	2.0	
$f(x, y)$	Production for match (x, y)	$0.6 + 0.4 (\sqrt{x} + \sqrt{y})^2$	Hagedorn et al. (2017)
β	Surplus division factor	0.72	Shimer (2005)
$m(\mathcal{U}, \mathcal{V})$	Matching function	$\kappa \mathcal{U}^\nu \mathcal{V}^{1-\nu}$	Hagedorn et al. (2017)
ν	Elasticity in meeting function	0.5	Hagedorn et al. (2017)
κ	Scale for meeting function	6.4	Unemployment rate
b	Worker unemployment benefit	1.0	Unemployment rate
c	Entry cost	2.40	Steady state $\mathcal{V}/\mathcal{U} = 1$
Steady State:			
$\bar{z}$	Steady state TFP	1	Shimer (2005)
$\bar{\delta}$	Steady state separation rate	0.2	BLS job tenure 5 years
$\bar{\varsigma}$	Steady state exogenous exit rate	0.01	No endogenous exit
Exogenous Aggregate Shock Process:			
A	TFP level	1.0	
$\varsigma_D, \varsigma_L, \varsigma_H$	Exogenous exit rates	0.63, 0.01, 0.01	No endogenous exit
δ_L, δ_H	Separation rates	0.22, 0.18	Shimer (2005)
$\delta_D(x, y)$	Separation at crisis state	0.23 to 3.9	Cajner et al. (2020)
λ_z	Poisson transition probabilities	0.4, 0.001, 0.0995	Shimer (2005)
$\sigma(z, \check{z})$	Exogenous exit at Poisson jumps	0 to 0.05	No endogenous exit
n_x	Discretization of worker types	5	Cajner et al. (2020)
n_y	Discretization of firm types	11	Cajner et al. (2020)



(a) Model with aggregate shock:
evaluation RMSE across the (x, y) grid



(b) Model without aggregate shock:
difference from conventional solution

Table of Contents

Methodology

Labor Search Model

Entry and Exit

On-The-Job Search Model

On-The-Job Search Model with Idiosyncratic Shocks

OTC Market

Non-transferable Utility (NTU)

Different Versions of Entry and Exit

1. Baseline (main text): free entry + exogenous exit at rate $\varsigma(z)$ and fraction $\sigma(z, \check{z})$ at z jumps, with full compensation for exit (g^f is a jump variable).
Calibrated s.t. ς, σ large enough $\Rightarrow$ no endogenous exit on equilibrium path.
2. General case: endogenous firm exit + partial compensation (fraction $\tau \in [0, 1]$ of lost surplus) (g^f jump variable). $\tau = 1$ & no endogenous exit $\Rightarrow$ baseline.
3. Short-lived vacancies (as in Lise-Robin '17; used in our OJS model): unfilled vacancies destroyed each instant, $V^v = 0$; no compensation; $\varsigma = \sigma = 0$.
4. Slow entry and exit (mass of firms $\mathcal{F}$ is a state variable).

1. Baseline: Free Entry, Exogenous Compensated Exit Back

- KFE: drift includes firm exit at rate $\varsigma(z)$; exogenous jump exit when z changes:

$$dg_t(x, y) = \mu^g(x, y, z_t, g_t) dt - \sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z})$$

$$\mu^g(x, y, z, g) = -(\delta(x, y, z) + \varsigma(z))g(x, y) + \frac{m(z, g)}{\mathcal{U}(z, g)\mathcal{V}(z, g)}\alpha(x, y, z, g)g^v(y)g^u(x)$$

- Master equation: no direct ς/σ loss terms (exit compensated); jump at post-jump dist.:

$$\begin{aligned} \rho S(x, y, z, g) = & zf(x, y) - \delta(x, y, z)S(x, y, z, g) + c - b \\ & - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g; S)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\ & - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g; S)} d\tilde{y} \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (S(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - S(x, y, z, g)) + D_g S \cdot \mu^g \end{aligned}$$

- Firm distribution g^f jumps to ensure free entry condition is satisfied:

$$0 = \mathbb{E}[V_t^v] = \int V^v(\tilde{y}, z, g) d\tilde{y}.$$

2. General Case: Endogenous Exit and Partial Compensation Back

- If free entry would require negative entry, firms are randomly forced to exit: at rate $\alpha_\mu^g(z, g)$ in the drift (indicator $\mathbf{1}_{\mu^F}$) or with prob. $\alpha_\sigma^g(z, \check{z}, g)$ at z jumps (indicator $\mathbf{1}_{\sigma^F}$), chosen so the free entry condition keeps holding.
- Exiting firms (workers) receive a transfer for fraction τ of their lost surplus.
- Master equation with endogenous exit and partial compensation:

$$\begin{aligned}
 \rho S(x, y, z, g) = & z f(x, y) - \delta(x, y, z) S(x, y, z, g) - (\varsigma(z) + \mathbf{1}_{\mu^F} \alpha_\mu^g)(1 - \tau) S(x, y, z, g) \\
 & + c - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g; S)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\
 & - b - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g; S)} d\tilde{y} \\
 & + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left\{ S(x, y, \check{z}, g + \gamma^g) - S(x, y, z, g) \right. \\
 & \left. - ((1 - \sigma) \mathbf{1}_{\sigma^F} \alpha_\sigma^g + \sigma)(1 - \tau) S(x, y, \check{z}, g + \gamma^g) \right\} + \langle D_g S, \mu^g \rangle
 \end{aligned}$$

where $\gamma^g = -(\sigma + \mathbf{1}_{\sigma^F} \alpha_\sigma^g)g$ is the jump in g ; μ^g includes exit rate $\varsigma + \mathbf{1}_{\mu^F} \alpha_\mu^g$.

- $\tau = 1$ and $\mathbf{1}_{\mu^F} = \mathbf{1}_{\sigma^F} = 0$ recovers the baseline master equation.

3. Slow Entry and Exit Back

- Vacancies long-lived; firms exit at exogenous rate ς (no jump exit, $\sigma = 0$); entrants draw type y from density $\bar{h}^f(y) = 1$; entry at rate $\psi \alpha^f$:

$$dg_t^f(y) = \underbrace{\psi \alpha^f(z, g, \mathcal{F}) \bar{h}^f(y)}_{\text{entry inflow}} dt - \underbrace{\varsigma g_t^f(y)}_{\text{exit outflow}} dt$$

$$\alpha^f(z, g, \mathcal{F}) = \mathbb{1} \left\{ \int V^v(\tilde{y}, z, g, \mathcal{F}) \bar{h}^f(\tilde{y}) d\tilde{y} > 0 \right\}$$

- Firm density stays uniform $\Rightarrow$ only the mass of firms $\mathcal{F}_t$ is an extra state:

$$d\mathcal{F}_t = \psi \alpha^f(z, g, \mathcal{F}) dt - \varsigma \mathcal{F}_t dt, \quad \mathcal{V}_t = \mathcal{F}_t - \mathcal{P}_t$$

- Surplus function has an additional state (and we need to solve for V^v): $S(x, y, z, g, \mathcal{F})$, with the master equation gaining the term $\partial_{\mathcal{F}} S(x, y, z, g, \mathcal{F}) \mu^{\mathcal{F}}(z, g, \mathcal{F})$.
- Limits: $\psi \rightarrow \infty$ recovers free entry (g^f jump variable); $\psi \rightarrow 0$ shuts down entry.

Table of Contents

Methodology

Labor Search Model

Entry and Exit

On-The-Job Search Model

On-The-Job Search Model with Idiosyncratic Shocks

OTC Market

Non-transferable Utility (NTU)

On-The-Job Search (OJS): Additional Environment Features

- Same worker types, firm types, and production function.
- Now all workers search; meetings $m(\mathcal{W}_t, \mathcal{V}_t)$; total search effort $\mathcal{W}_t := \mathcal{U}_t + \phi \mathcal{E}_t$.
Meeting rates: $\mathcal{M}_t^u = \frac{m_t}{\mathcal{W}_t}$ (unemployed), $\mathcal{M}_t^e = \phi \frac{m_t}{\mathcal{W}_t}$ (employed), $\mathcal{M}_t^v = \frac{m_t}{\mathcal{V}_t}$ (vacancy).
- Vacant $\tilde{y}$ -firm meets unemployed x -worker: Nash bargaining with worker weight β .
- Poaching firm $\tilde{y}$ meets worker in (x, y) match: [parameterized bargaining protocol](#):
 - Worker moves iff $S_t(x, \tilde{y}) \geq S_t(x, y) \geq 0$; gets share β of incremental surplus $S_t(x, \tilde{y}) - S_t(x, y)$... and share $\chi \in [0, 1]$ of the incumbent firm's surplus.
 - Nests: Bertrand competition ($\chi = 1, \beta = 0$; PVR '02), sequential auctions ($\chi = 1, \beta > 0$; CPVR '06), incremental Nash bargaining ($\chi = 0$; Pissarides '94).
 - χ does not enter the surplus master equation (only wages) $\Rightarrow$ same S for all χ .
- Endogenous separation at rate η when $S_t(x, y) < 0$: $\alpha_t^b(x, y) = 1 - \alpha_t(x, y)$.
- Short-lived vacancies (Lise-Robin '17): unfilled vacancies destroyed each instant $\Rightarrow V^v = 0$; free entry on the flow value of a vacancy; $\varsigma = \sigma = 0$.

Recursive Characterization For Equilibrium Surplus (OJS)

- Can characterize equilibrium with the master equation for the surplus:

$$\begin{aligned} \rho S(x, y, z, g) = & z f(x, y) - b - (\delta(x, y, z) + \eta \alpha^b(x, y, z, g)) S(x, y, z, g) \\ & + \mathcal{M}^e \int \alpha^e(x, y, \tilde{y}, z, g) \beta(S(x, \tilde{y}, z, g) - S(x, y, z, g)) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\ & - \mathcal{M}^u \int \alpha(x, \tilde{y}, z, g) \beta S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (S(x, y, \tilde{z}, g) - S(x, y, z, g)) + D_g S(x, y, z, g) \cdot \mu^g(z, g) \end{aligned}$$

where the on-the-job acceptance and endogenous breakup indicators are:

$$\alpha^e(x, y, \tilde{y}, z, g) := \mathbb{1}\{S(x, \tilde{y}, z, g) \geq S(x, y, z, g) \text{ and } S(x, \tilde{y}, z, g) \geq 0\}, \quad \alpha^b = 1 - \alpha$$

- Simpler than baseline: $V^v = 0$ (short-lived vacancies) $\Rightarrow$ no firm opportunity cost term.
 $\beta = 0$ kills both g -dependent terms $\Rightarrow$ block recursivity as in Lise-Robin '17.

OJS: Free Entry Condition

- The flow value of creating a vacancy y : rents from hiring unemployed workers and poaching employed workers ($\mathcal{C}^u = \frac{\mathcal{U}}{\mathcal{W}}$, $\mathcal{C}^e = \frac{\phi \mathcal{E}}{\mathcal{W}}$ prob. met worker is u/e):

$$\begin{aligned} J^v(y, z, g) = & \mathcal{M}^v \mathcal{C}^u \int \alpha(\tilde{x}, y, z, g) (1 - \beta) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\ & + \mathcal{M}^v \mathcal{C}^e \int \int \alpha^e(\tilde{x}, \tilde{y}, y) (1 - \beta) [S(\tilde{x}, y, z, g) - S(\tilde{x}, \tilde{y}, z, g)] \frac{g(\tilde{x}, \tilde{y})}{\mathcal{E}} d\tilde{x} d\tilde{y} \end{aligned}$$

- Free entry condition (entrants draw y from uniform density $\bar{h}^v(\tilde{y}) = 1$):

$$0 = -c + \int_0^1 J^v(\tilde{y}, z, g) \bar{h}^v(\tilde{y}) d\tilde{y}$$

- With homothetic $m(\mathcal{W}, \mathcal{V})$, invert to solve $\mathcal{V}$ as a function of (z, g) and $S(\cdot, \cdot, z, g)$.

On-the-job-search: KFE

- The KFE becomes:

$$\begin{aligned}
 dg_t(x, y) = & \underbrace{-(\delta(x, y, z) + \eta \alpha^b(x, y, z_t, g_t)) g_t(x, y) dt}_{\text{Outflow: exogenous \& endogenous separation}} \\
 & - \underbrace{\mathcal{M}_t^e \int \alpha^e(x, y, \tilde{y}, z_t, g_t) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} g_t(x, y) dt}_{\text{Outflow: workers poached to better matches}} \\
 & + \underbrace{\mathcal{M}_t^u \alpha(x, y, z_t, g_t) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) dt}_{\text{Inflow: unemployed workers hired}} \\
 & + \underbrace{\mathcal{M}_t^e \int \alpha^e(x, \tilde{y}, y, z_t, g_t) \frac{g_t^v(y)}{\mathcal{V}_t} g_t(x, \tilde{y}) d\tilde{y} dt}_{\text{Inflow: workers poached from other matches}} =: \mu^g(x, y, z_t, g_t) dt
 \end{aligned}$$

Q4. Block recursive model impact on IRF to negative TFP shock?

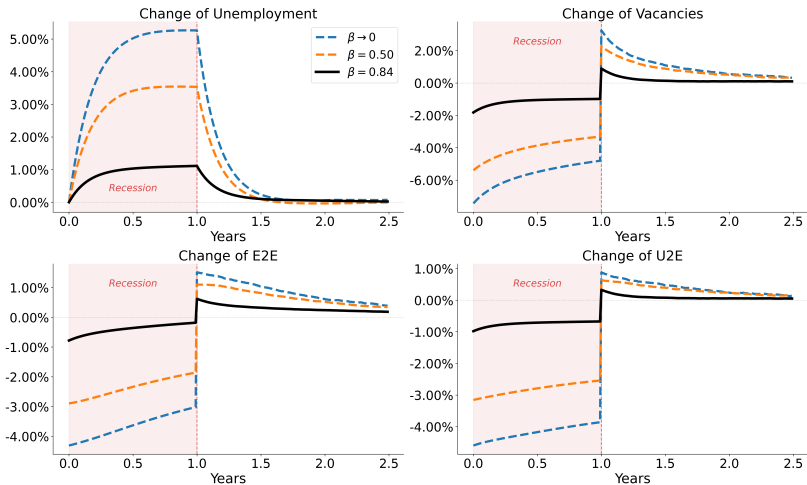


Table of Contents

Methodology

Labor Search Model

Entry and Exit

On-The-Job Search Model

On-The-Job Search Model with Idiosyncratic Shocks

OTC Market

Non-transferable Utility (NTU)

On-The-Job Search Model with Idiosyncratic Shocks

- Introduce match specific (idiosyncratic) productivity shock ϵ (Mortensen and Pissarides, 1994) that follows a continuous time Markov chain with intensities $\lambda(\epsilon, \epsilon')$.
- Production output from matches $F(x, y, \epsilon, z) = A_z A_\epsilon f(x, y)$; new matches draw initial $\epsilon \sim \pi(\epsilon)$.
- **Endogenous separation**: agents immediately break up the match when a new draw of ϵ makes the surplus negative ($\alpha^b(x, y, \epsilon, z, g) = \mathbb{1}\{S(x, y, \epsilon, z, g) < 0\}$).

On-The-Job Search Model with Idiosyncratic Shocks: Master Equation

$$\begin{aligned}
 \rho S(x, y, \epsilon, z, g) &= \rho(P(x, y, \epsilon, z, g) - V^u(x, z, g)), & V^v &= 0 \text{ (short-lived vacancies)} \\
 &= F(x, y, \epsilon, z) - \delta(x, y, z)S(x, y, \epsilon, z, g) - b \\
 &+ \mathcal{M}^e \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) \beta(S(x, \tilde{y}, \tilde{\epsilon}, z, g) - S(x, y, \epsilon, z, g)) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\
 &- \mathcal{M}^u \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha(x, \tilde{y}, \tilde{\epsilon}, z, g) \beta S(x, \tilde{y}, \tilde{\epsilon}, z, g) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\
 &+ \sum_{\check{\epsilon} \neq \epsilon} \lambda(\epsilon, \check{\epsilon}) ((1 - \alpha^b(x, y, \check{\epsilon}, z, g))(S(x, y, \check{\epsilon}, z, g) - S(x, y, \epsilon, z, g)) \\
 &\quad - \alpha^b(x, y, \check{\epsilon}, z, g) S(x, y, \epsilon, z, g)) \\
 &+ \sum_{\check{z} \neq z} \lambda(z, \check{z}) (S(x, y, \epsilon, \check{z}, g) - S(x, y, \epsilon, z, g)) + \langle D_g S, \mu^g \rangle
 \end{aligned}$$

where $\alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) = \mathbb{1}\{S(x, \tilde{y}, \tilde{\epsilon}, z, g) \geq S(x, y, \epsilon, z, g) \text{ \& } S(x, \tilde{y}, \tilde{\epsilon}, z, g) \geq 0\}$.

On-The-Job Search Model with Idiosyncratic Shocks: KFE

$$\begin{aligned}
 dg_t(x, y, \epsilon) = & -\delta(x, y, z) g_t(x, y, \epsilon) dt \\
 & - \underbrace{\underbrace{\mathcal{M}_t^e}_{\text{Rate. e-worker meets}} \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \underbrace{\alpha_t^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon})}_{\text{Prob. accept}} \underbrace{\frac{g_t^v(\tilde{y})}{\mathcal{V}_t}}_{\text{Prob. meet } \tilde{y}} d\tilde{y}}_{\text{Mass accepting poaching offer}} \underbrace{g_t(x, y, \epsilon)}_{\text{Mass at } (x, y, \epsilon)} dt \\
 & + \underbrace{\mathcal{M}_t^u}_{\text{Rate. u-worker meets}} \pi(\epsilon) \alpha_t(x, y, \epsilon) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) dt \\
 & + \mathcal{M}_t^e \pi(\epsilon) \sum_{\tilde{\epsilon}} \int \alpha_t^e(x, \tilde{y}, \tilde{\epsilon}, y, \epsilon) \frac{g_t^v(y)}{\mathcal{V}_t} g_t(x, \tilde{y}, \tilde{\epsilon}) d\tilde{y} dt \\
 & - \sum_{\check{\epsilon} \neq \epsilon} \lambda(\epsilon, \check{\epsilon}) g_t(x, y, \epsilon) dt + \sum_{\check{\epsilon} \neq \epsilon} \lambda(\check{\epsilon}, \epsilon) (1 - \alpha_t^b(x, y, \epsilon)) g_t(x, y, \check{\epsilon}) dt
 \end{aligned}$$

(Last line: outflow when ϵ switches away; inflow from $(x, y, \check{\epsilon})$ when $\check{\epsilon}$ switches to ϵ , net of endogenous breakups.)

On-The-Job Search Model with Idiosyncratic Shocks: Results

- Recession experiment ($z = z_L$ for one year): cumulative net outflows from matches rise more, and recover more slowly, at the low idiosyncratic state ϵ_L (left)
...because low- ϵ matches carry lower surplus and sit near the breakup threshold $S = 0$.
- Endogenous separation flow $Q(x, y, z, g_{\text{erg}})$: recessions raise endogenous separations relatively more for high- (x, y) matches (right).

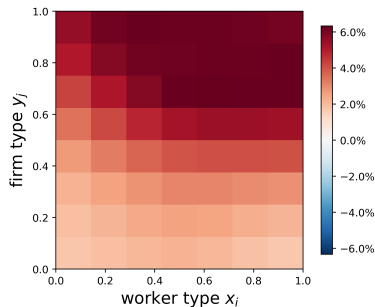
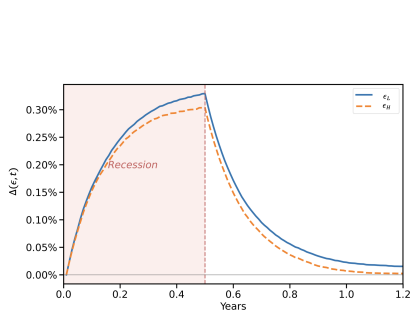


Table of Contents

Methodology

Labor Search Model

Entry and Exit

On-The-Job Search Model

On-The-Job Search Model with Idiosyncratic Shocks

OTC Market

Non-transferable Utility (NTU)

Environment: Setting, Bonds, and Households

- Continuous time, infinite horizon environment.
- There are many bonds, $k \in \{1, \dots, K\}$, in positive net supply s_k :
 - Every bond pays the same dividend $\delta > 0$.
 - Bond k matures at rate $1/\tau_k$ (average maturity τ_k).
- Populated by a unit-mass continuum of infinitely-lived and risk-neutral investors:
 - An investor can hold either zero or one share of at most one type of asset.
 - Investor type $j \in \{1, \dots, J\}$ gets flow utility $\delta - \psi(j, k)$ from holding bond k .
 - Agents **switch type** from i to j at rate $\lambda_{i,j}$.
- Aggregate (default) state $z \in \{z_1, \dots, z_n\}$, switches at rate $\zeta_{z,z'}$.
At state z , asset k pays a fraction $\phi(k, z)$ of the coupon and the principal.

Distribution and Bargaining

- An investor's state is made up of her holding cost $j \in \{1, \dots, J\}$ and her ownership status, for each asset type $k \in \{1, \dots, K\}$ (owner o or non-owner n). Hence the set of investor idiosyncratic states is:

$$A = \{1n, 2n, \dots, Jn, 1o1, \dots, 1oK, 2o1, \dots, 2oK, Jo1, \dots, JoK\} \quad (3)$$

- The rate of contact between investors with states a and b is:

$$\mathcal{M}_{a,b} = \kappa_{a,b} g_a g_b \quad (4)$$

- Agents a, b **trade with each other**, engage in generalized Nash bargaining with bargaining power $\beta_{a,b}$.

Value Functions for Non-Owners and Owners [back](#)

- Value function for non-owner with type i satisfies:

$$\begin{aligned}\rho_i V(in, g, z) = & \sum_a \kappa_{in,a} \alpha(in, a, g, z) \beta_{in,a} S(in, a, z, g) \\ & + \sum_k \xi_{i,k} (V(iok, g, z) - V(in, g, z)) + \sum_{j \neq i} \lambda_{i,j} (V(jn, g, z) - V(in, g, z)) \\ & + \sum_{z'} \zeta_{z,z'} (V(in, g, z') - V(in, g, z)) + \sum_{a \in A} \partial_{g_a} V(in, g, z) \mu^g(a, z)\end{aligned}$$

- Value function for an investor of type i holding asset k satisfies:

$$\begin{aligned}\rho_i V(iok, g, z) = & \delta \phi(k, z) - \psi(i, k) + \frac{1}{\tau_k} (V(in, g, z) + \pi(k, z) - V(iok, g, z)) \\ & + \sum_a \kappa_{iok,a} \alpha(iok, a, g, z) g_a \beta_{iok,a} S(iok, a, g, z) + \sum_{j \neq i} \lambda_{i,j} (V(jok, g, z) - V(iok, g, z)) \\ & + \sum_{z'} \zeta_{z,z'} (V(iok, g, z') - V(iok, g, z)) + \sum_{a \in A} \partial_{g_a} V(iok, g, z) \mu^g(a, z).\end{aligned}$$

$\alpha(in, jok, g, z)$: indicator for whether the trade is accepted upon matching.

Endogenous Yield Curve and Impulse Responses [back](#)

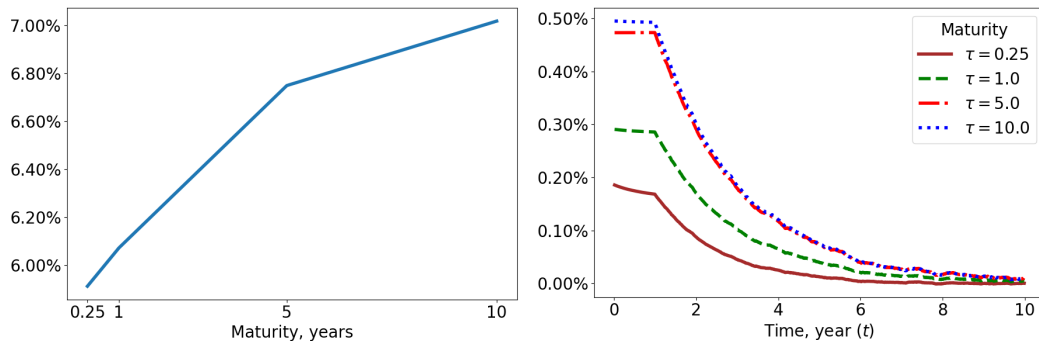


Figure: Yield curve at ergodic steady state and impulse responses. Right figure: proportional bond yield change compared to the ergodic yield at each maturity following a one-year recession.

Table of Contents

Methodology

Labor Search Model

Entry and Exit

On-The-Job Search Model

On-The-Job Search Model with Idiosyncratic Shocks

OTC Market

Non-transferable Utility (NTU)

Non-transferable Utility (NTU)

- Main change compared to the benchmark model: for a match (x, y) , agent x receives payoff $zf(x, y)$ and agent y receives $zf(y, x)$. No surplus to be split via bargaining.
- Computational implication:
 1. We no longer solve a single master equation for the match surplus.
 2. Instead, we directly approximate the value functions $V^u(x, z, g)$, $V^e(x, y, z, g)$, $V^v(y, z, g)$, and $V^p(x, y, z, g)$ using four neural networks, trained jointly to minimize a weighted sum of residuals from the four master equations.

Non-transferable Utility (NTU, Ct'd)

- Mean squared loss as a function of type in the master equations of V^u (left) and V^e (right)

